

COMPSCI 3AC3 - Assignment 1

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Contents

Question 1

Show that:

(a) $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Solution:

(b) $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{Z}/2\mathbb{Z}$.

Solution:

Question 2

Show that:

(a) $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$.

Solution:

(b) $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{Z}/2\mathbb{Z}$.

Solution:

Question 3

Show that:

(a) For every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution:

(b) For every $\alpha \in \mathbb{Z}/2\mathbb{Z}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution:

Question 4

Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$